



Software Manual: American Sign Language Recognition for Emergency Alerts: A Machine Learning Approach

Project Name: American Sign Language Recognition for Emergency Alerts: A Machine Learning Approach

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Authors: Ritika Rani & Kashish Haryani

Institute: SZABIST University Larkana Campus

Coordinator: Mr. Abdul Khaliq

Supervisor: Mr. Muhammad Bux Soomro

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1- Introduction

The **American Sign Language Recognition for Emergency Alerts** project utilizes machine learning techniques to identify American Sign Language (ASL) gestures in real time and convert them into text and audio format. This software is aimed at helping individuals with hearing and speech impairments communicate during emergencies by recognizing ASL gestures and providing output based on the recognized signs.

Audience

This manual is intended for:

- **End-Users:** Those who will use the software to communicate emergency alerts using ASL.
- **Developers:** Those who may need to modify or maintain the software.
- **Administrators:** Users responsible for installation and configuration.

Scope

This manual covers the installation, usage, and maintenance of the software application. It does not cover detailed machine learning theory or deep algorithm explanations, as these are outside the scope of this guide.

2- System Requirements

Hardware Requirements

- **Processor:** Intel i5 or higher
- **RAM:** Minimum 4 GB
- **Storage:** Minimum 100 MB free space for installation
- **Camera:** A webcam or compatible camera for real-time feed capture
- **Graphics:** Integrated graphics or dedicated GPU for machine learning processing

Software Requirements

- **Operating System:** Windows 10 or higher
- **Python:** Python 3.8 or higher
- **Required Libraries:**
 - OpenCV
 - NumPy
 - Matplotlib / Seaborn
 - MediaPipe

- pyttsx3 (for text-to-speech functionality)
 - PIL (Python Image Library)
 - Scikit-learn (for machine learning)
 - Tkinter for GUI
- **IDE:** Visual Studio Code (VS Code)

Network Requirements

- Stable internet connection for downloading libraries and dependencies

3- Installation Instructions

Step 1: Installing Python

- If you don't have Python installed, download and install the latest version of Python from the [official Python website](https://www.python.org/).
- During installation, make sure to check the box "**Add Python to PATH**".

Step 2: Setting Up the Development Environment

1. Open **Visual Studio Code**.
2. Open the project folder containing the source code.

Step 3: Install Dependencies

- Open a terminal in VS Code.
- Install required libraries by running the following commands:

```

pip install opencv-python
pip install numpy
pip install matplotlib
pip install pyttsx3
pip install pillow
pip install scikit-learn
pip install mediapipe
pip install Tkinter
pip install seaborn

```

Step 4: Running the Software

1. Once the installation is complete, navigate to the main project file.
2. Open the terminal and run:

```
python Main.py
```

Or you can directly run the project from the play sign on the top right corner of the screen.

The software will launch the splash screen and proceed to the main interface.

4- Getting Started

First Steps

1. **Launching the Application:**
 - To run the application, execute **Main.py** from the terminal or through VS Code's run feature.
 - The application will open with a splash screen.
2. **Basic Configuration:**
 - No complex setup is required from the user. The software will automatically detect the camera and start recognizing ASL gestures once the application is opened.

5- Features and Functions

Splash Screen

Description: The splash screen appears when the software is launched. It displays the logo and project title.

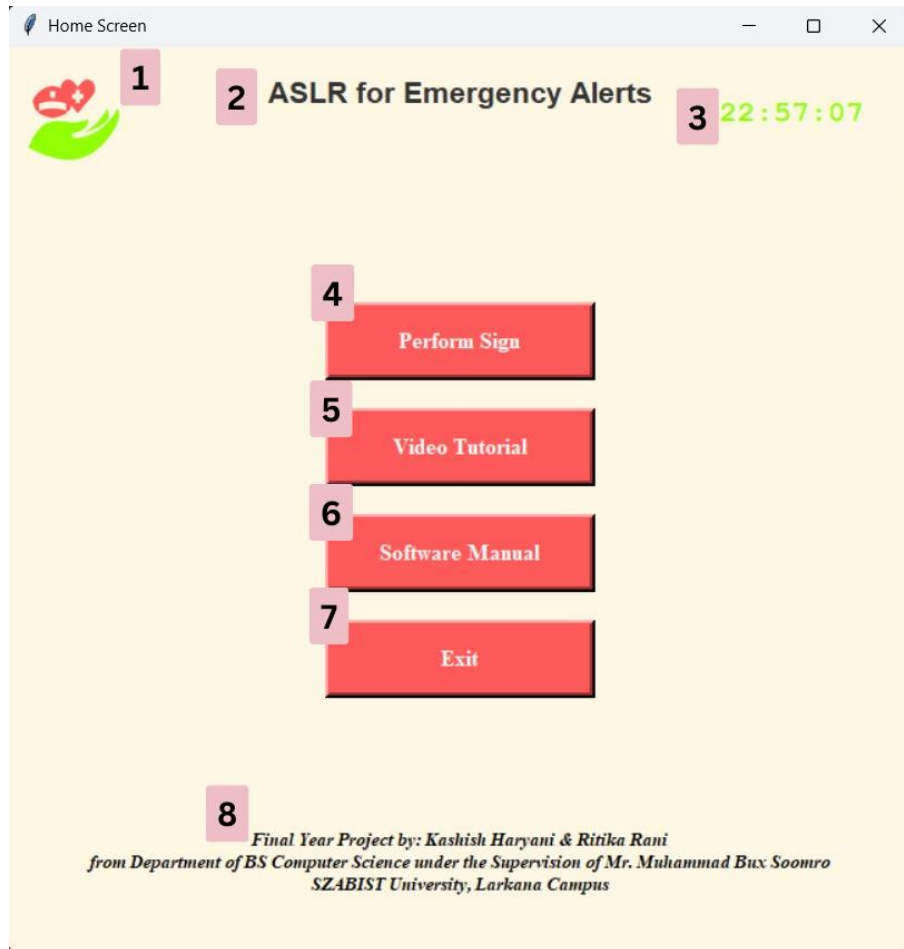


Functionality: The splash screen serves as a welcome page and provides brief details about the project. After a few seconds, it transitions to the main screen.

- 1- Logo of the software
- 2- Name of the project

Main Screen

Description: This is the central interface where users interact with the software.



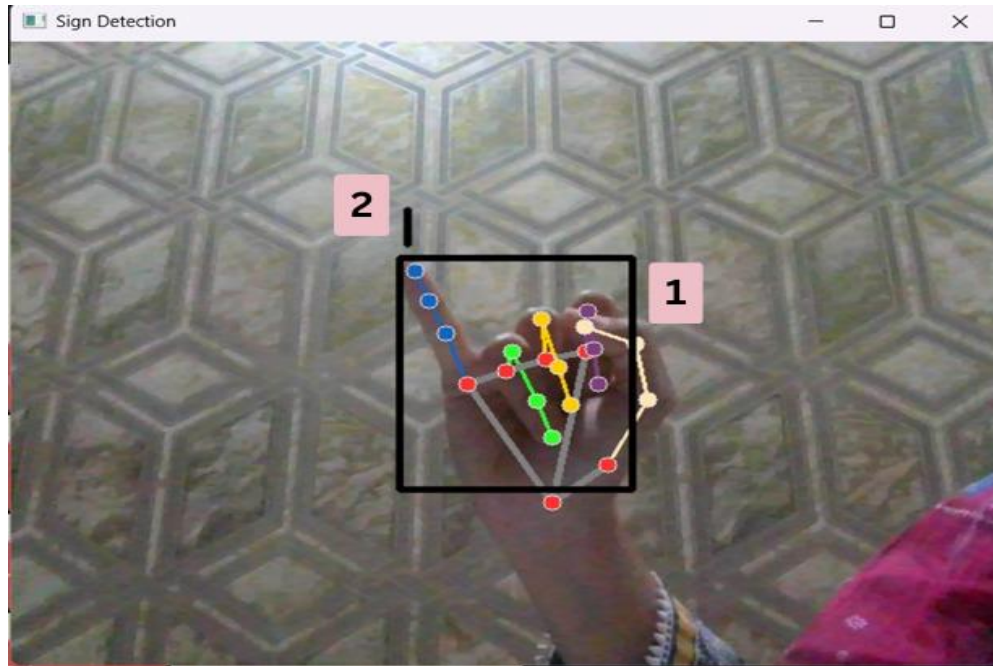
Functionality:

- 1- **Image box:** Box containing logo image.
- 2- **Text Label:** Text label for Project name.
- 3- **Digital Clock:** Clock to view time.
- 4- **Button1: “Perform Sign”** Button to trigger the camera window where user will perform the signs.
- 5- **Button2: “Video Tutorial”** It will display the tutorial of how to operate the software in video format.
- 6- **Button3: “Software Manual”** It will open the pdf file of the user manual.
- 7- **Button4: “Exit”** It will terminate the software upon clicking on the yes option displayed on the screen right after pressing the “Exit” button.
- 8- **Text Label:** This label contains the credits of the project.

Camera Window

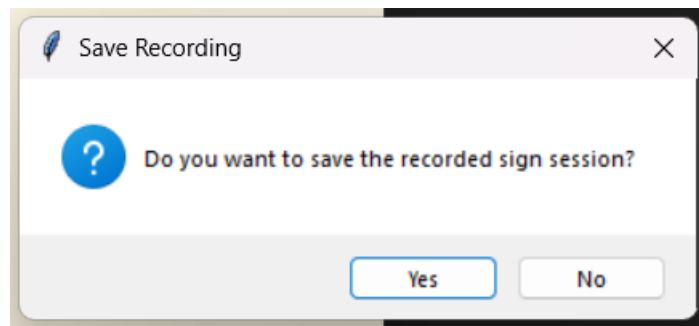
Description: The camera feed interface uses your computer’s camera to capture hand gestures.

Additionally the performed signs by the user will be recorded in video format and user will be asked if they want to save the recorded session or not.



Functionality:

- 1- The camera will continuously stream the video to identify ASL signs.
- 2- Once emergency sign is detected, the text and audio output will be generated.
- 3- The user can close the camera window using ESC key.
- 4- The session will be recorded once the camera is triggered, if user wants to save the recorded session they will get confirmation message
- 5- The system will save the recorded video in the 'ASLR_Output_Recordings' folder for future references if the user clicks "Yes".
- 6- If the user doesn't want the recorded session to be saved, they will simply click on "No" and the recorded video will be discarded.



6- Troubleshooting

Common Issues

Issue: Camera not detected.

Solution: Ensure that your webcam is connected properly. Check camera permissions and ensure no other applications are using it.

Issue: ASL recognition is inaccurate.

Solution: Ensure the lighting is adequate for gesture recognition. Adjust your hand positions for better accuracy and perform the signs within the vicinity of the camera.

7- FAQs

Q1: What if the software does not recognize the signs properly?

- Ensure that your hand gestures are within the camera frame, and check the lighting conditions.

Q1: How much distance between the user and camera?

- The user should perform the sign within the vicinity of the camera.

8- Contact Information

Contact Number: +92 0000000000

E-mail: support@gmail.com

9- Glossary

ASL (American Sign Language): A visual language used by the Deaf and Hard-of-Hearing communities.

Machine Learning: A type of artificial intelligence that allows systems to learn from data and make decisions without being explicitly programmed.

Gesture Recognition: The process of interpreting hand gestures as input to trigger actions.